

Edward Han

778-927-1538 | edwardhan20020806@gmail.com | linkedin.com/in/edwardhan86

EDUCATION

The University of British Columbia

Vancouver, BC

Bachelor of Applied Science in Computer Engineering

Expected Graduation November 2025

- **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Computer Architecture, Embedded Systems Design, Machine Learning, Software Engineering, Database Systems, Human-Computer Interaction, Digital Logic Design

EXPERIENCE

Software Developer

January 2025 - February 2026

Skydweller Technologies inc.

Vancouver, BC

- Developed UI using Python for a 3D modeling application (software under NDA) that automates drone flight-path planning using intelligent mesh processing and real-time visualization, making flight planning over 10x faster (over 90% reduction).
- Created comprehensive mesh preprocessing pipeline featuring spatial and remeshing algorithms for large building-scale meshes.
- Implemented an export pipeline supporting ArduPilot Mission Planner/ QGroundControl, DJI FlightHub 2 / Pilot 2, and Gazebo (SDF) formats, featuring a GIS-accurate local-to-GPS conversion system using the pyproj library.

PROJECTS

Workout Form Optimizer | UBC

January 2026 - Present

- Led a 7-person team (first-year through PhD) to build the project's core CV stack: MediaPipe pose, joint-angle features, and exercise-specific form classifiers for squat, deadlift, and bench press.
- Integrated FastAPI upload and streaming with the browser client, implemented the main endpoints for per-frame angles and form feedback, and coordinated contributors across CV, backend, and frontend.
- Architected modular angle utilities and swappable pose backends, implemented OpenCV video analysis paths, wrote classifier tests, and mentored contributors via design review and onboarding

IntelliDish - AI Powered Recipe App for Android | UBC

January 2025 - April 2025

- Developed a full-stack Android app using Android Studio and the OpenAI API to generate personalized recipes based on user preferences
- Ensured code reliability through testing with Jest, Espresso, Codacy, and structured code reviews.
- Automated the CI/CD pipeline with GitHub Actions for continuous integration and deployment
- Implemented SSL handshake for secure communication and deployed the app on AWS EC2 using Docker containers

OS/161 - Operation System Kernel Development | UBC

September 2024 - December 2024

- Implemented core operating system features in OS/161 using C and assembly, including system calls, process management, and virtual memory
- Developed file I/O functionality and contributed to a basic virtual file system
- Implemented synchronization primitives to ensure thread-safe kernel operations and solve multi-threading problems

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript/TypeScript, Kotlin, HTML/CSS, Assembly, Verilog/SystemVerilog, GLSL/HLSL, SQL

Frameworks: React, Node.js, Jest, Espresso

Developer Tools: Git, Docker, GitHub Actions, Android Studio, Blender, AWS EC2, Isaac Sim, Isaac Lab, Quartus, ModelSim, Gazebo, ArduPilot SITL Simulator, Figma, Codacy, ChampSim

Libraries: pyproj, bpy, bmesh, WebGL, pandas, NumPy, Matplotlib

Platforms/ Hardware: DE1-SoC FPGA, ArduPilot, QGroundControl, DJI Pilot 2, DJI FlightHub 2

Interfaces/ Protocols: UART, IIC, SPI, CAN